

Smart Modular Workstation for Disaster Relief

Submission for Problem Statement 4

May 19, 2025

Submitted by:

Akula Uday Kiran (1RV23AS004)

Kiran V Airani (1RV23AS020)

Shanthosh K V (1RV23AS053)

Tejas L (1RV23AS061)

Department of Aerospace Engineering

Contents

1	Executive Summary	2
2	Module Descriptions	2
2.1	Module 1: Power and Communication Unit	2
2.2	Module 2: Medical and Sustenance Unit	6
2.3	Module 3: Survival Kit	7
2.4	Module 4: Deployment Mechanism	10
2.4.1	Final Assembly	13
3	Key Features and Design Considerations	14
4	Computational Analysis	14
4.1	Engineering Calculations	14
4.2	CFD Analysis of Descent Phase	16
4.3	Static Structural Analysis	17
4.3.1	One corner impact	17
4.3.2	Two corner impact	17
4.3.3	All corner impact	18
5	Mission Use Case Description	19
6	Conclusion	19

1 Executive Summary

This report outlines the design and development of a Smart Modular Workstation intended for use in disaster relief scenarios. Conceived as a portable, foldable, and modular system, the workstation is engineered for rapid deployment in emergency situations such as floods and earthquakes. It supports critical operations including medical assistance, drone coordination, and communication infrastructure.

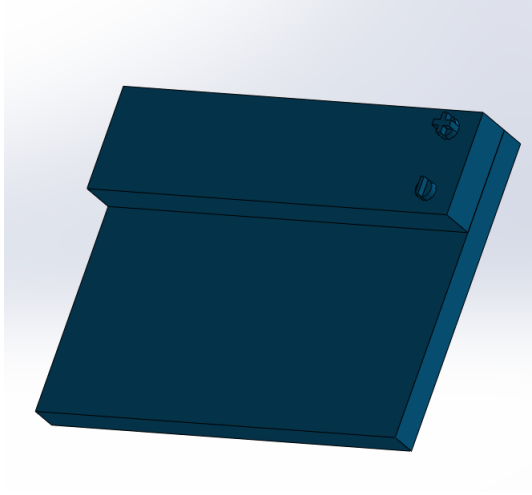
The workstation is divided into four functional modules, each equipped with integrated solar panels, energy storage systems, and communication components. Key features include a power and communication hub, a medical and sustenance module, a survival kit, and a specialized deployment mechanism. The system incorporates sustainable design principles and advanced technology such as Raspberry Pi-based IoT monitoring and a compact wastewater purification setup. This document satisfies the requirements of both preliminary CAD layout submission (Round 1) and integrated systems evaluation (Round 2), with additional engineering analyses and operational details.

2 Module Descriptions

2.1 Module 1: Power and Communication Unit

This core module provides electrical power and long-range communication capabilities. Positioned at the base for optimal weight distribution, it comprises:

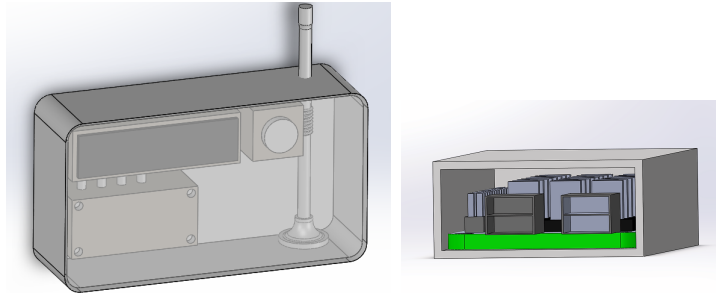
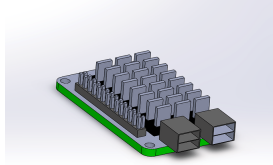
- **Battery System:** Includes integrated charging/discharging circuitry. A Lithium-Polymer battery has been chosen for its high energy density and flexible form factor.



Estimated weight: 30.71 kg.

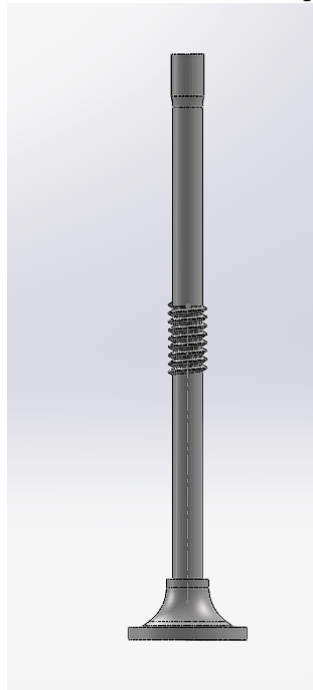
The battery system has an estimated energy capacity of 3,685 to 4,914 Wh, based on an energy density of 150-200 Wh/kg for the battery cells and accounting for the battery management system. With an average power consumption of 100 W (considering the Raspberry Pi, IoT dashboard, communication systems, and intermittent use of other electronics), the endurance ranges from 36.85 to 49.14 hours (approximately 1.5 to 2 days). During a drop at 20 m/s, the battery is protected by the descent system's airbags, ensuring no significant capacity loss upon impact.

- **IoT-Enabled Raspberry Pi:** Facilitates real-time monitoring of power status, GPS location, and occupancy data through a dashboard interface.

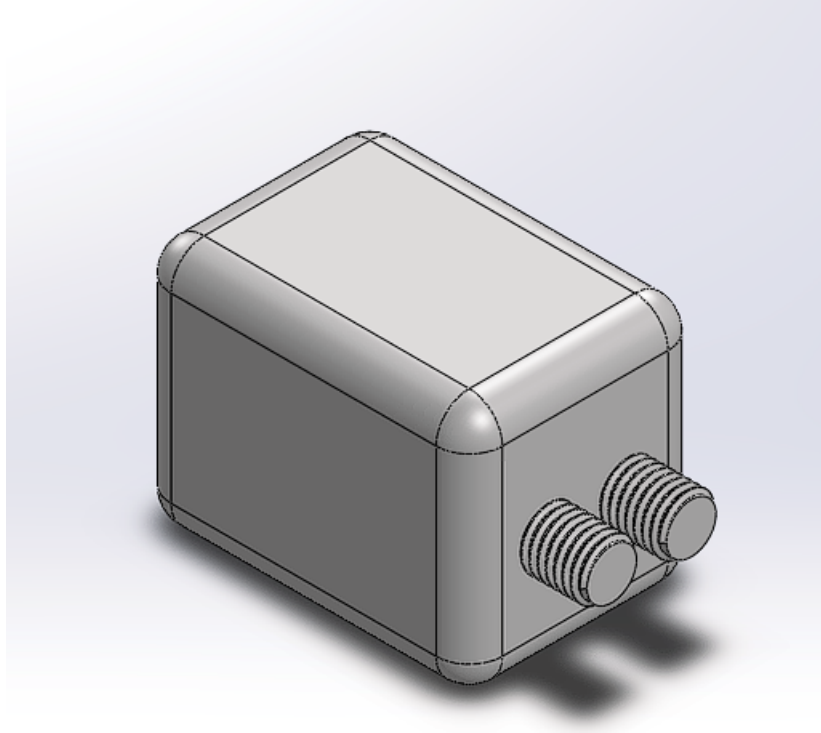


The Raspberry Pi 5 weighs 50 grams, and along with casing 74 grams.

- **Antenna and Ham Radio System:** Enables robust long-distance communication, crucial in disaster response operations.

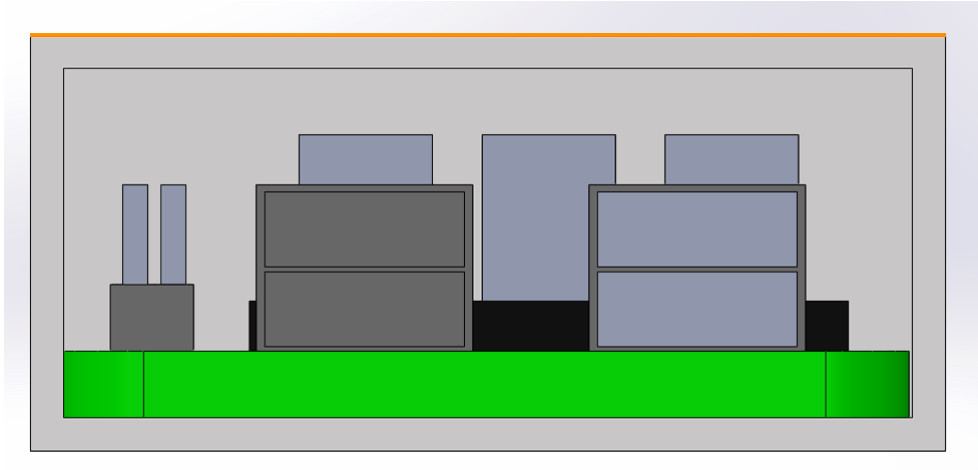


Material: PET plastic with copper wiring. Weight: 15 grams.

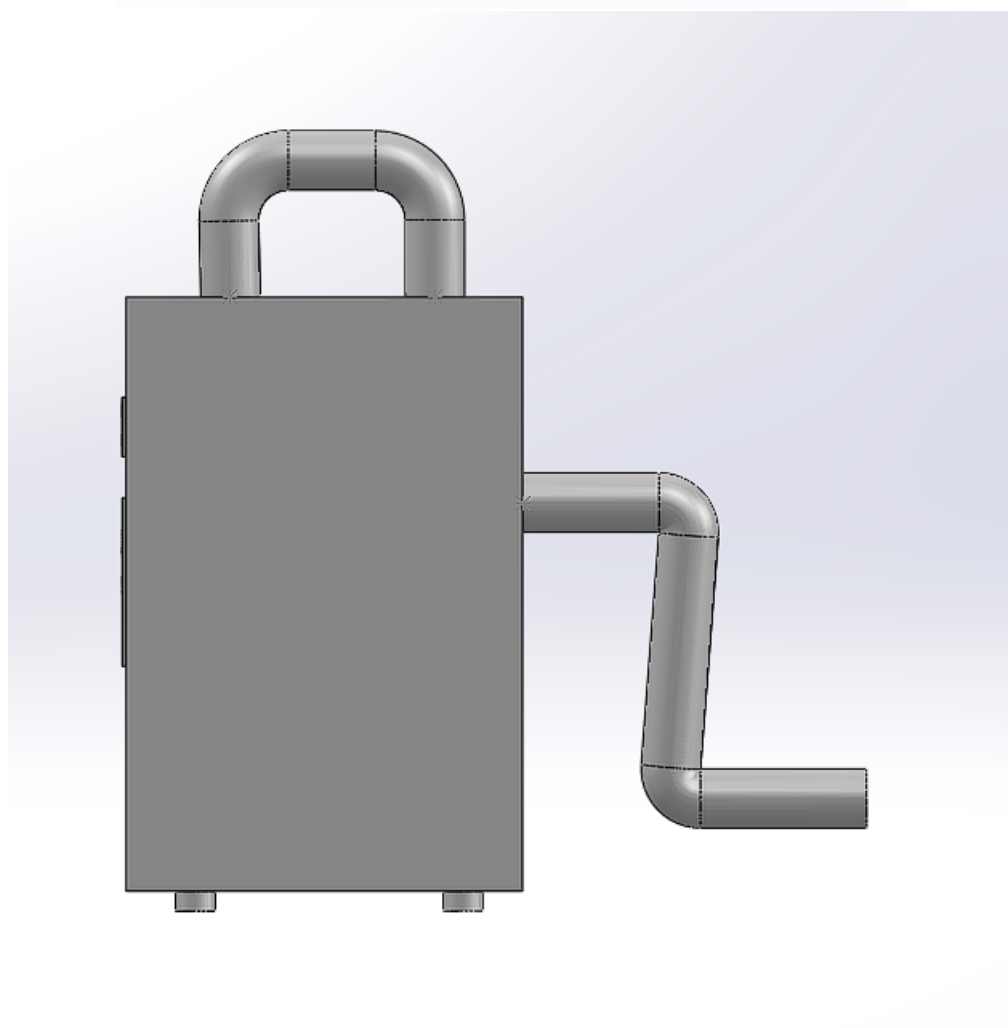
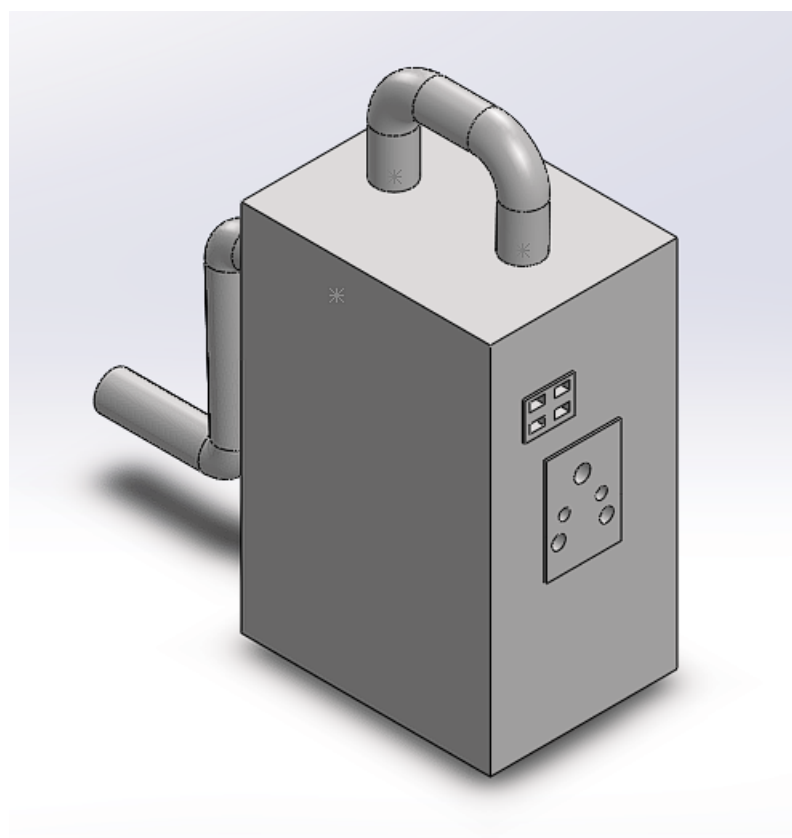


Ham radio housing: PET plastic; weight approximately 730 grams.

- **I/O Ports:** Equipped with four USB interfaces via the Raspberry Pi 5.



- **Manual Generator:** A hand-crank power generator ensures emergency energy availability.



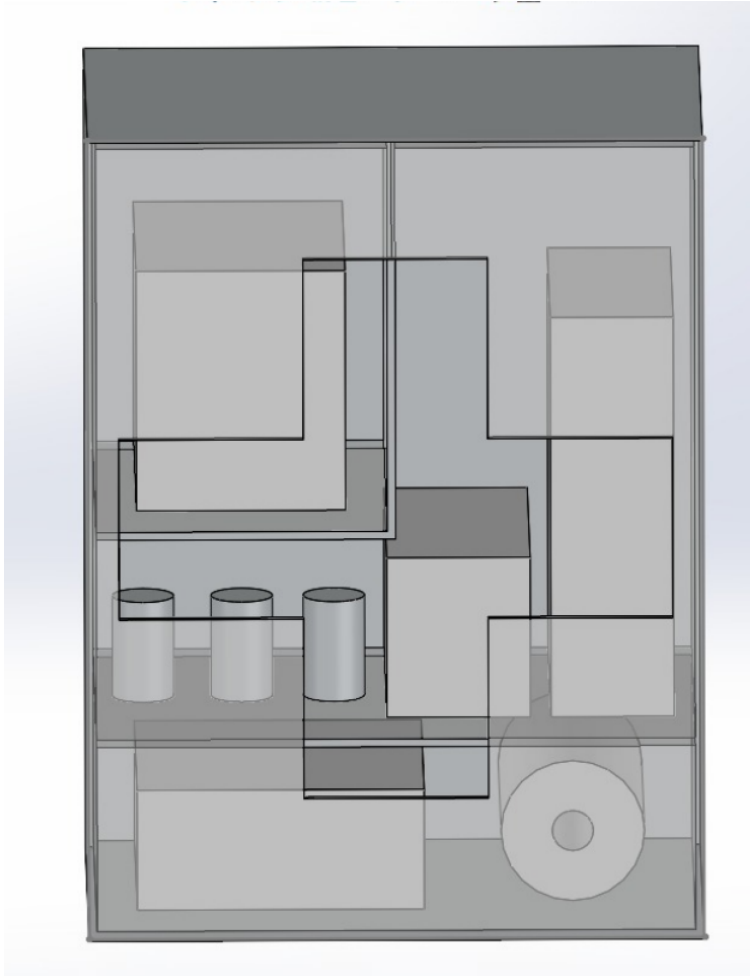
Assembly primarily composed of PET plastic. Weight: approximately 13.3 kg.

- **Satellite Connectivity:** Provides GPS and intermittent satellite communication for tracking and navigation.

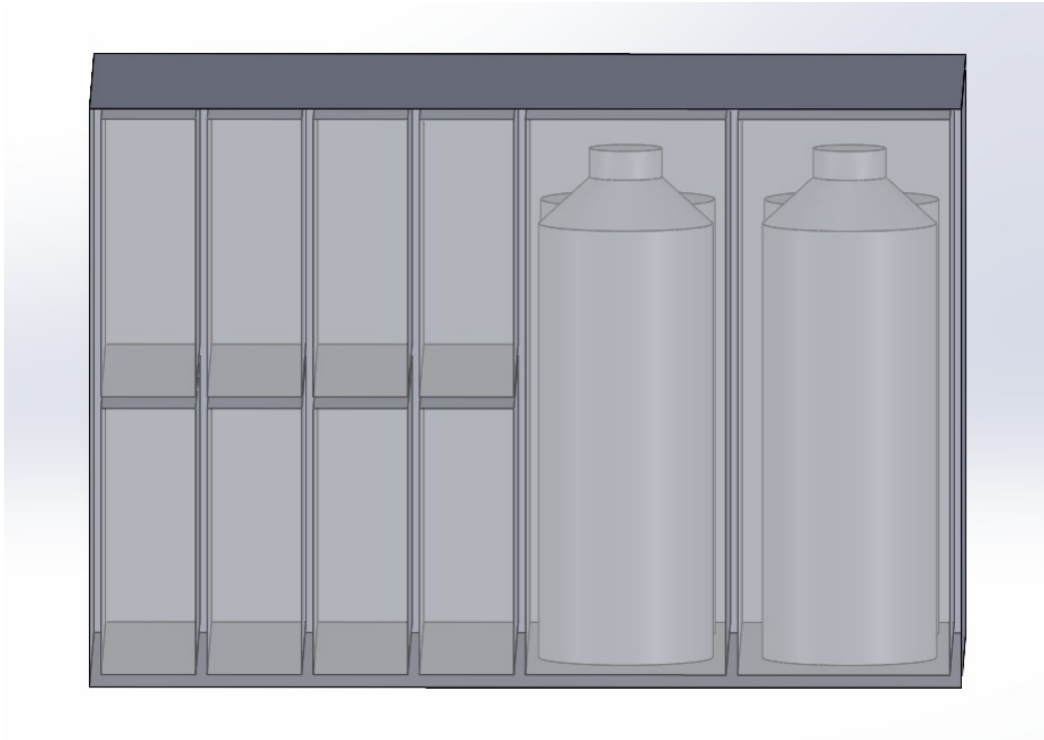
2.2 Module 2: Medical and Sustenance Unit

This module addresses essential health and nutritional requirements:

- **Medical Supplies:** Stocked with first-aid and emergency response materials.

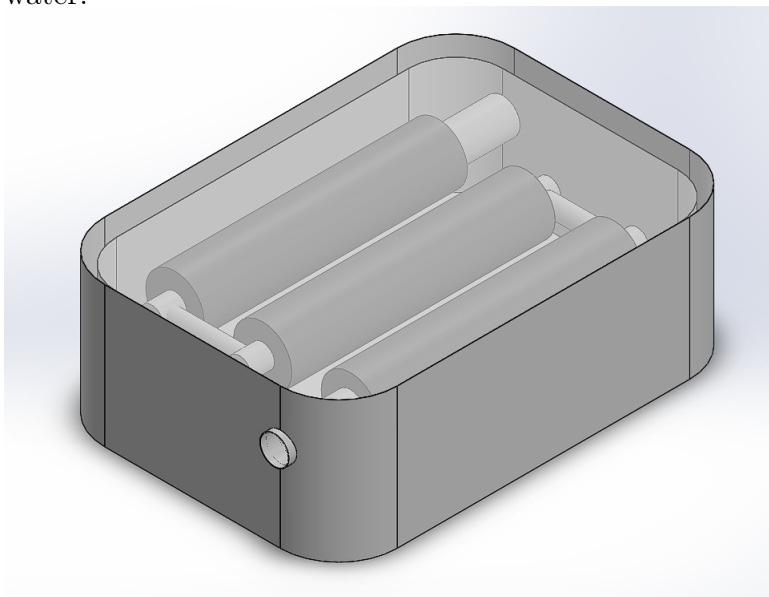


- **Dehydrated Food Packs:** Long-shelf-life rations stored in compact PET compartments.



Compartment weight: approximately 441 grams.

- **Water Purification System:** Filters contaminated water to produce potable water.



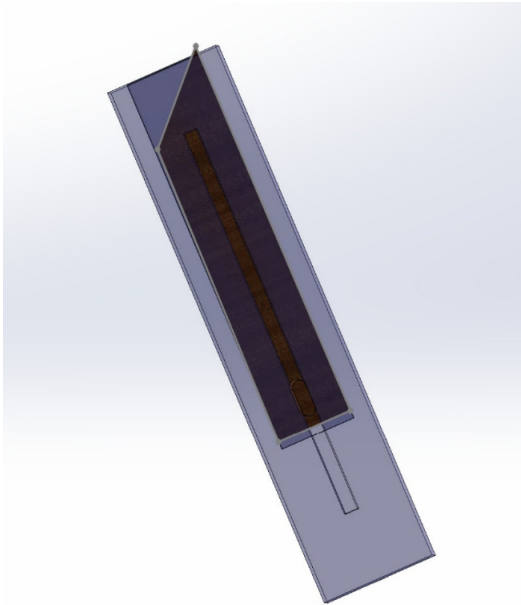
Constructed from PET plastic; dry weight: 2756 grams.

2.3 Module 3: Survival Kit

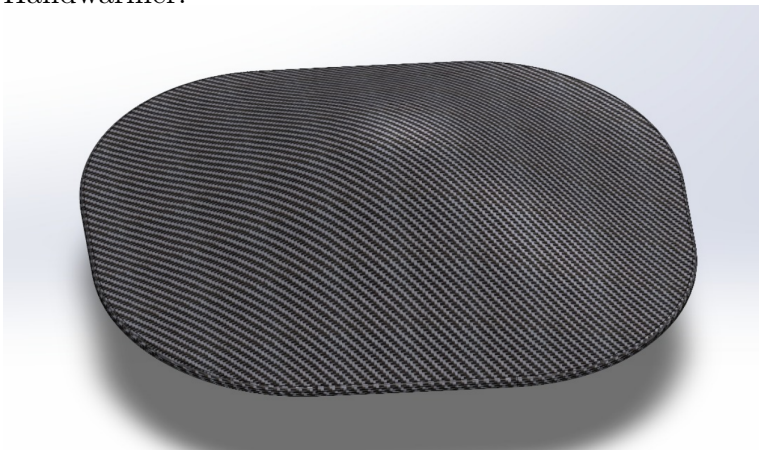
This module enhances field survival and reconnaissance capabilities:

- **Essential Tools:** Includes utility knives, ropes, fire starters, torches, flare guns, life jackets, and protective masks.

Knife:



Handwarmer:

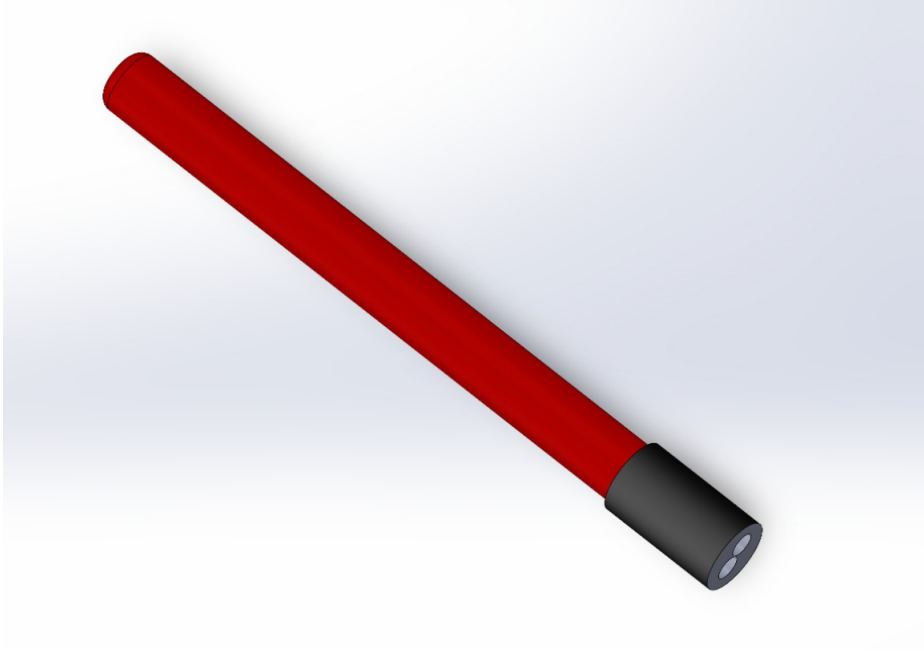


Rope:

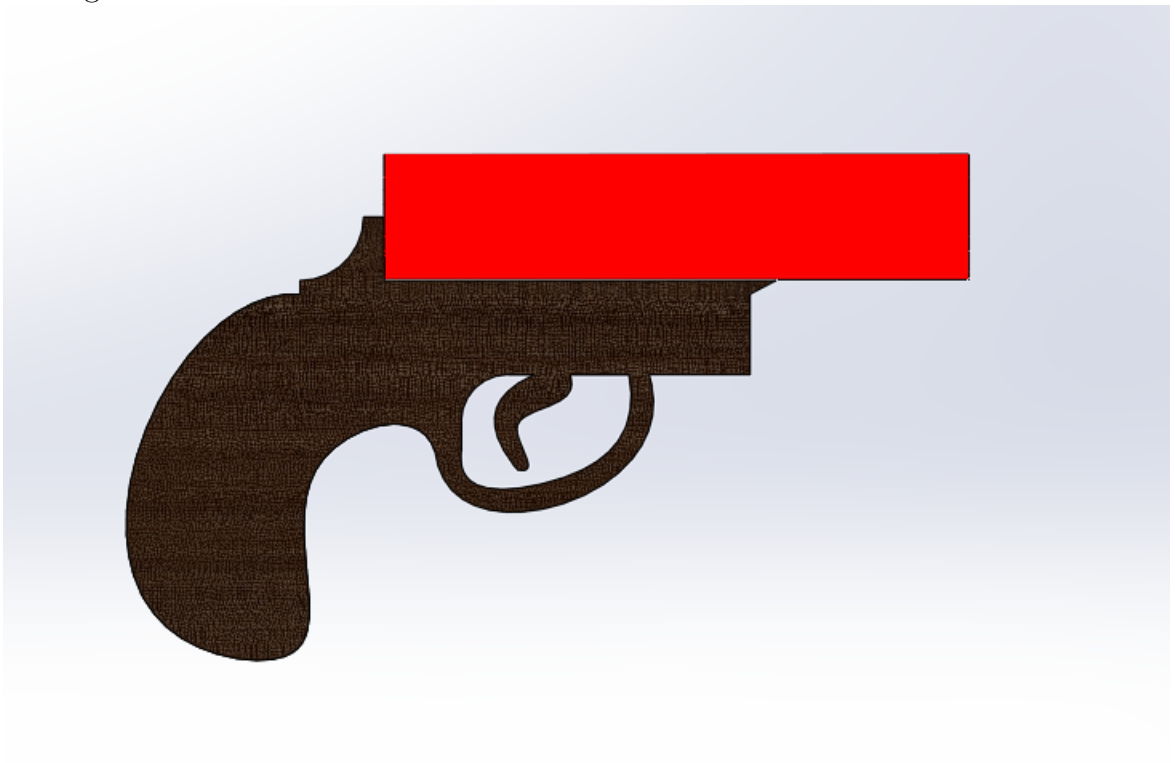


Rope material: Neoprene. Weight: 70 grams.

Glow stick:

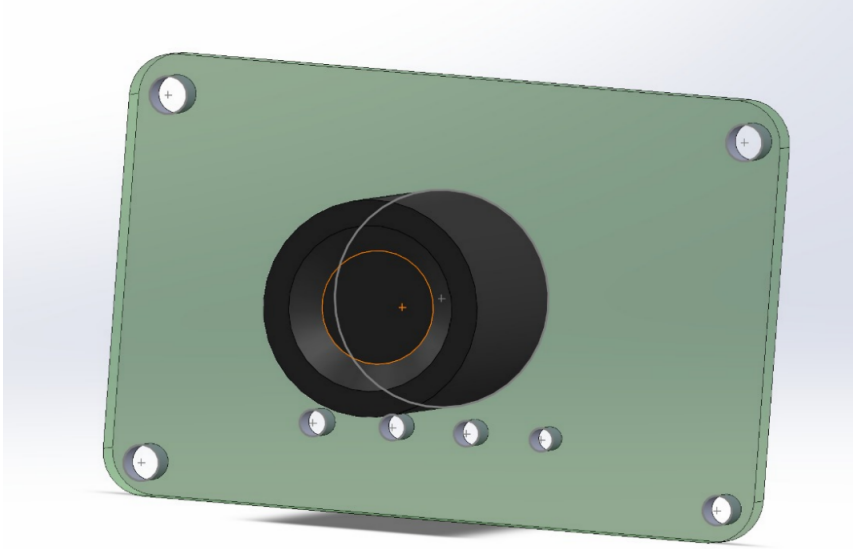


Flare gun:



Assuming the gun is made up of Titanium alloy, it weighs 540 grams.

- **Thermal Imaging:** An infrared heat-sensing camera aids in locating survivors.



This module weighs 300 grams.

2.4 Module 4: Deployment Mechanism

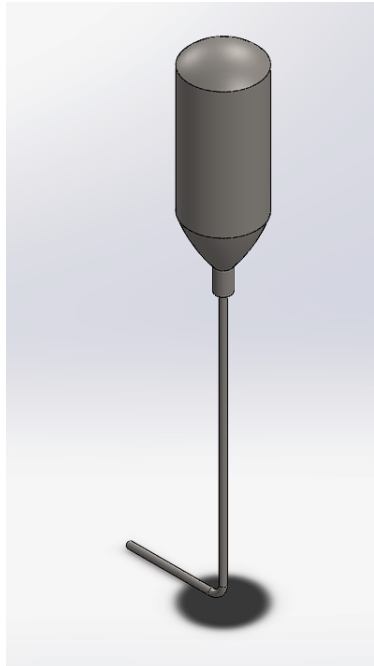
This module ensures safe delivery and operational readiness:

- **Descent System:** Incorporates automotive airbag mechanisms, parachutes, CO2 canisters, and inflatable cushions for impact mitigation. The workstation is delivered via air-drop from a helicopter or large drone. During descent, the Raspberry Pi, equipped with GPS and IMU sensors, provides real-time navigation data to guide the workstation to the designated drop zone. The parachute system controls the descent rate, reducing the terminal velocity to approximately 18 m/s, while CO2 canisters and airbags manage the final landing phase, protecting the payload at velocities up to 20 m/s. The workstation itself serves as the primary payload, with internal modules secured to withstand impact forces. Post-mission, the system can be disassembled manually and retrieved by ground vehicles or aircraft, leveraging its modular design for efficient packing.

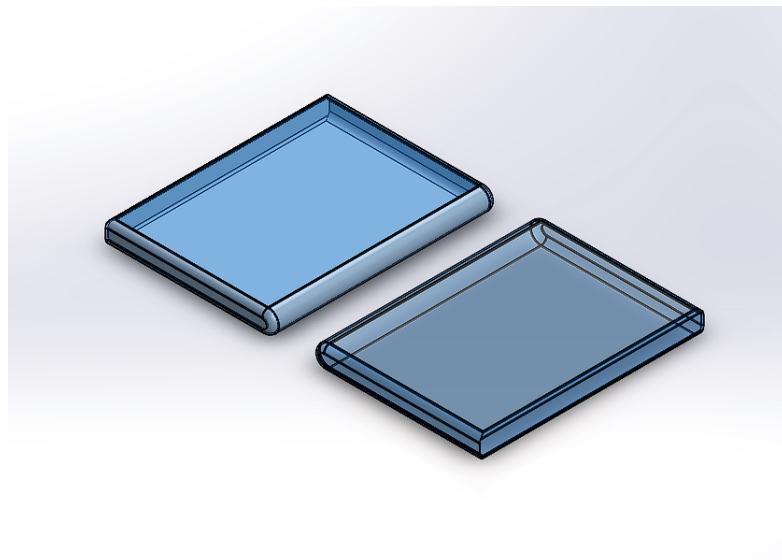
Parachute:



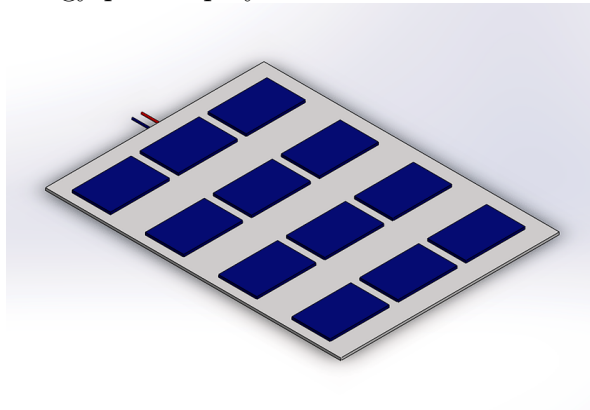
CO2 Cannisters:



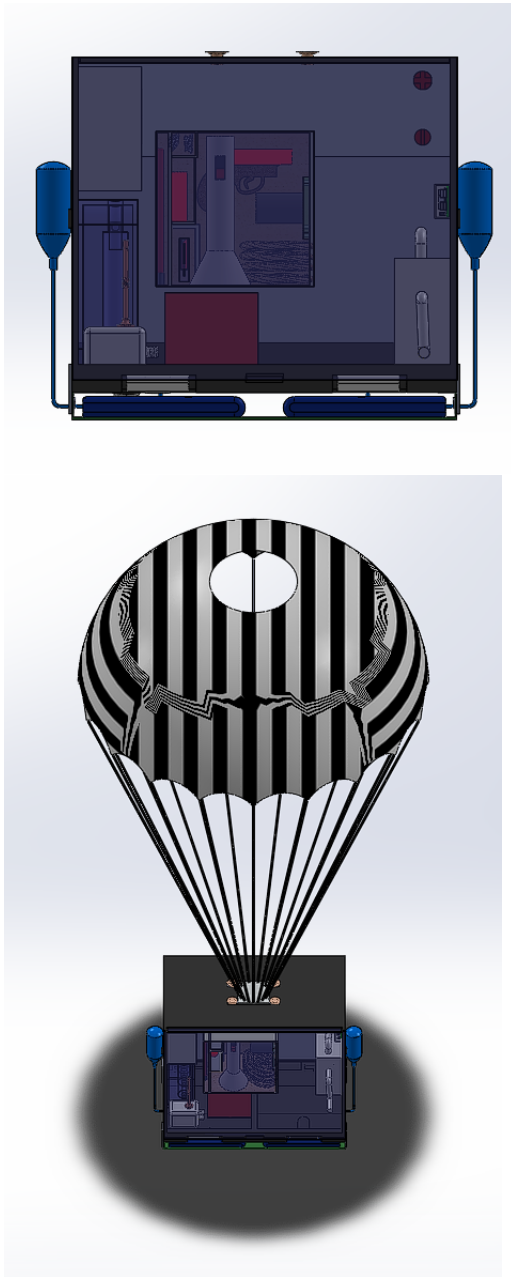
Airbags:

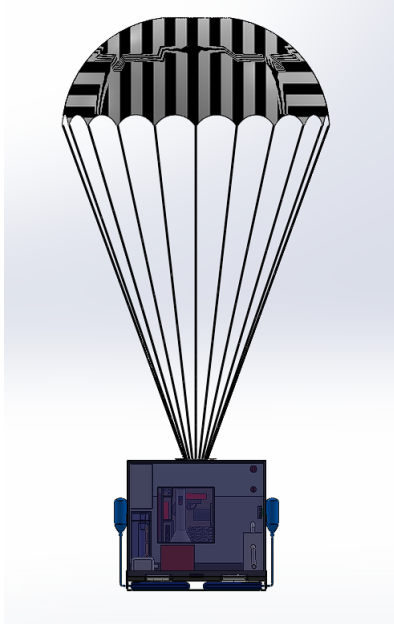


- **Solar Panel Array:** Foldable, lightweight solar panels provide sustainable energy post-deployment.



2.4.1 Final Assembly





3 Key Features and Design Considerations

- **Modular Architecture:** Facilitates ease of transport, maintenance, and reconfiguration.
- **Structural Stability:** Balanced design with low center of gravity achieved by battery placement.
- **Hybrid Power System:** Utilizes both solar and manual energy sources for redundancy.
- **Real-Time Telemetry:** IoT dashboard monitors operational parameters in real time.
- **Robust Engineering:** Finite element simulations validate mechanical endurance under deployment stress.
- **Operating Environment:** The workstation is designed to operate in diverse disaster environments, including flood zones, earthquake-damaged areas, and regions affected by extreme weather. It functions within a temperature range of -10°C to 40°C, withstanding high humidity, potential water submersion, dust, and debris. Materials such as PET plastic and aluminum provide corrosion resistance and durability, while the low center of gravity ensures stability on uneven terrain.

4 Computational Analysis

4.1 Engineering Calculations

The terminal velocity of the workstation during descent was calculated using the formula:

$$v_t = \sqrt{\frac{2mg}{\rho AC_d}}$$

where:

- $w = 75 \text{ kg}$ (estimated total mass of the workstation, including all modules),
- $g = 9.81 \text{ m/s}^2$ (acceleration due to gravity),
- $\rho = 1.225 \text{ kg/m}^3$ (air density at sea level),
- $A = 0.217 \text{ m}^2$ (cross-sectional area without parachute),
- $C_d = 0.6$ (drag coefficient without parachute).

This yields:

$$v_t = \sqrt{\frac{2 \times 75}{1.225 \times 0.217 \times 0.6}} = \sqrt{\frac{150}{0.159495}} \approx 30.667 \text{ m/s}$$

With the parachute deployed, assuming $C_d = 1.75$ and $A = 1.8237 \text{ m}^2$ (effective parachute area), the terminal velocity becomes:

$$v_t = \sqrt{\frac{2 \times 75}{1.225 \times 1.8237 \times 1.75}} = \sqrt{\frac{150}{3.90955}} \approx 6.1941 \text{ m/s}$$

This aligns with the CFD simulation result of 6 m/s, confirming the descent system's efficacy.

For a dropping velocity of 10 m/s (e.g., a controlled descent scenario), the impact force upon landing is estimated assuming a stopping distance of 0.5 m provided by the airbags:

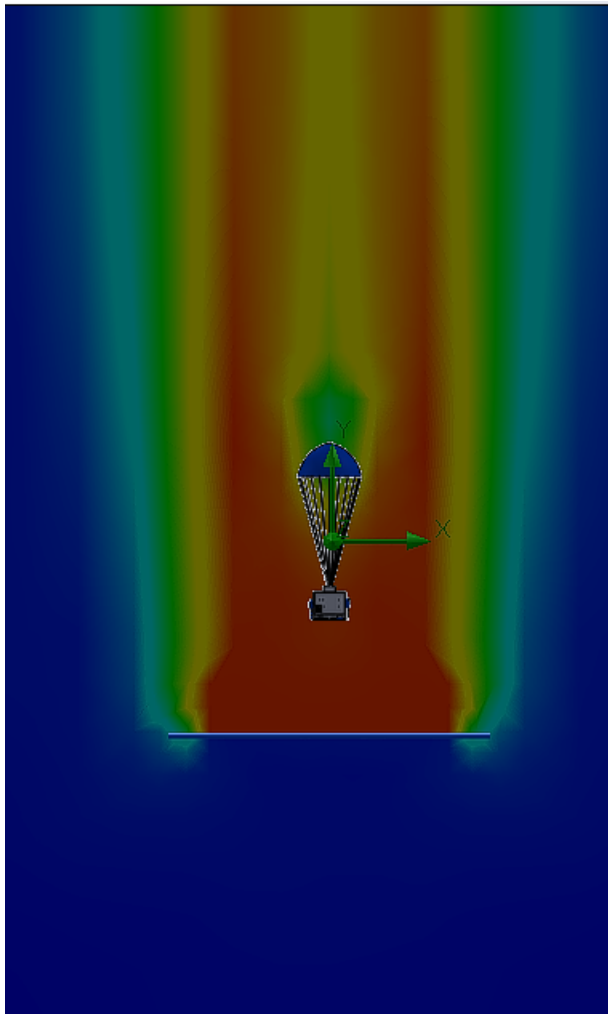
$$a = \frac{v^2}{2d} = \frac{10^2}{2 \times 0.5} = \frac{100}{1} = 100 \text{ m/s}^2$$

$$F = ma = 75 \times 100 = 7500 \text{ N}$$

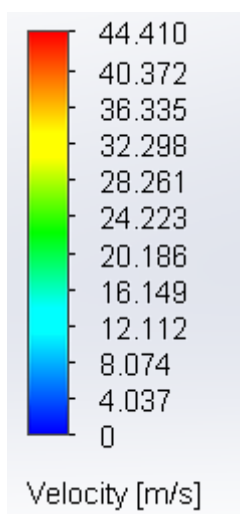
This force is within the structural tolerances of the workstation, supported by the airbags and inflatable cushions, ensuring operational integrity post-landing.

4.2 CFD Analysis of Descent Phase

The terminal velocity during descent was estimated to be 31 m/s based on physical modeling. This served as the input for a CFD simulation:



Legend:



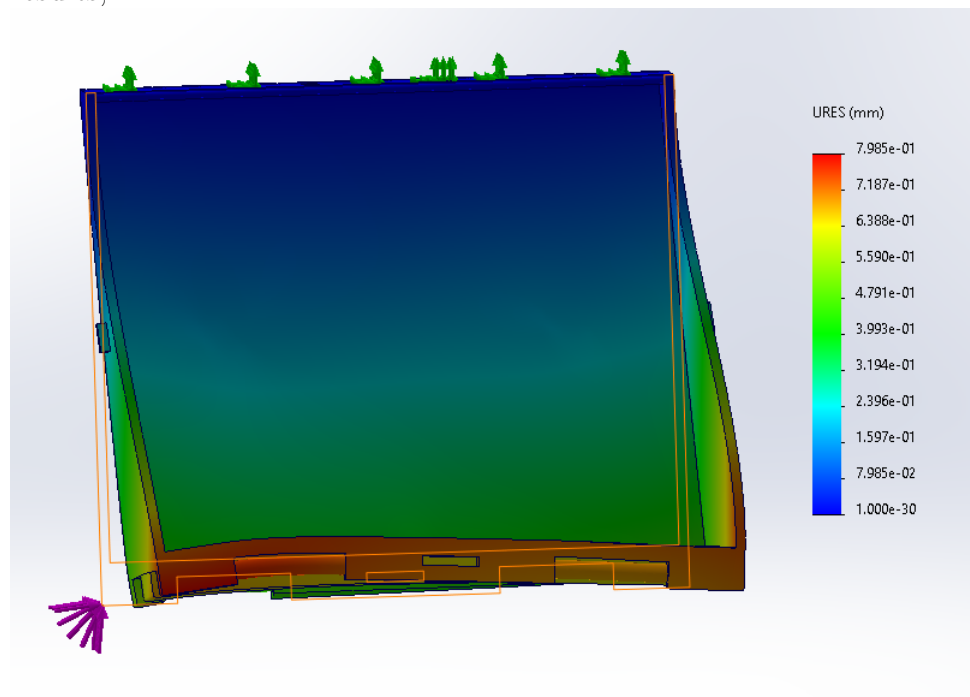
Simulation results indicate that the parachute system reduces terminal velocity to approximately 6.2 m/s. In combination with the airbag mechanism, this configuration effectively mitigates impact forces, ensuring the integrity of the equipment upon landing.

4.3 Static Structural Analysis

For this analysis, we have assumed that the impact will mainly occur at the corners. All ideal cases. The impact force is taken to be 30000N, while the actual value is 7500N, this gives a Factor of Safety of 4

4.3.1 One corner impact

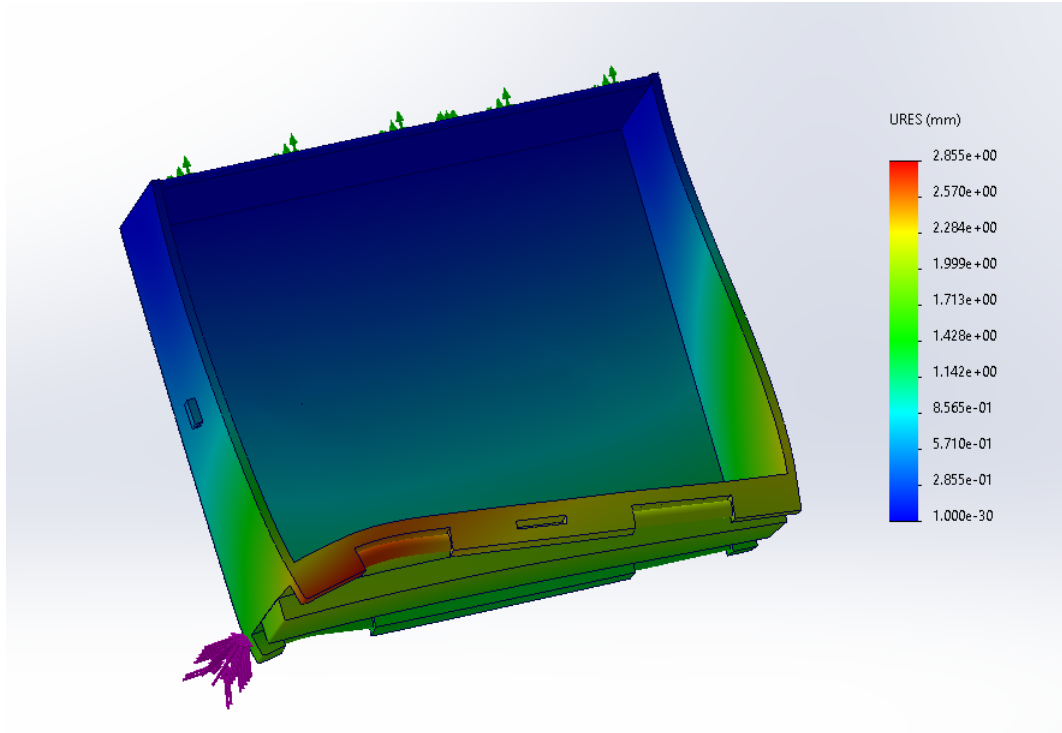
This is an ideal case, the body is assumed to fall on only one of the corner. Here are the results,



In this case, the maximum deformation is estimated to be around 0.7985mm, which is in limits of deformation.

4.3.2 Two corner impact

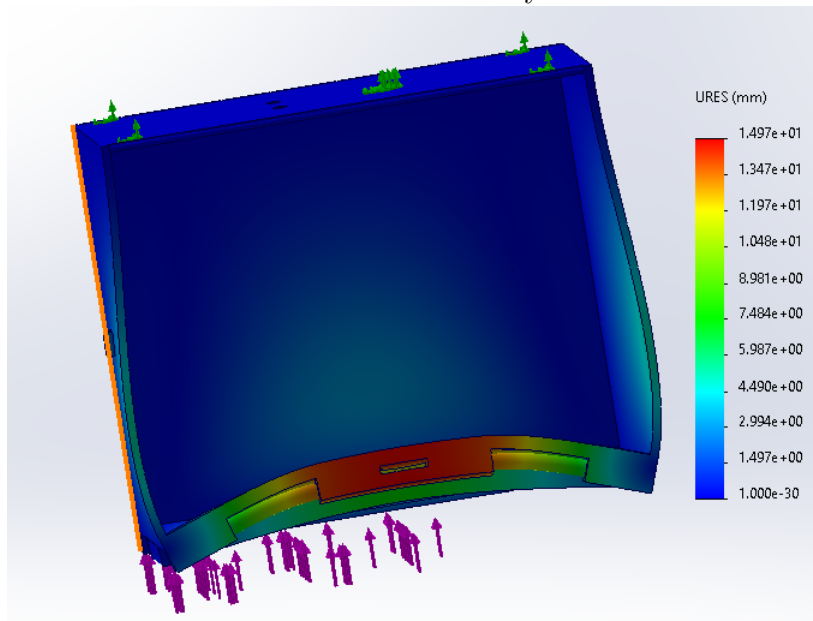
This is an ideal case in which the body is assumed to fall on 2 corners,



In this case, the maximum deformation is estimated to be around 1.829mm, which is also in limits.

4.3.3 All corner impact

This is an ideal case in which the body is assumed to fall on all 4 corners,



In this case, the maximum deformation is estimated to be around 1mm, which is also in limits.

5 Mission Use Case Description

In the aftermath of a major natural disaster in a remote region, communication infrastructure is disabled and local resources are severely strained. A Smart Modular Workstation is air-dropped into the affected zone from a helicopter at an altitude of 1000 meters. During its descent, an onboard navigation system ensures accurate targeting of the designated drop site. The parachute slows its fall to 18 m/s, while airbags cushion the landing, allowing for safe deployment even at speeds up to 20 m/s.

Upon touchdown, the workstation automatically activates. Solar panels deploy to harness renewable energy, and the power module establishes a satellite connection, broadcasting real-time GPS coordinates and enabling emergency communication via ham radio.

Relief teams utilize the workstation as a mobile command center. Through its integrated IoT dashboard, they coordinate drone-based aerial assessments to identify affected areas and locate survivors. Medical supplies stored in Module 2 are distributed to those in need, while its built-in water purification system ensures access to safe drinking water. Module 3 is equipped with a thermal imaging camera, aiding in the detection of individuals trapped or hidden from view, such as under rubble or foliage.

Designed for rapid response, the Smart Modular Workstation plays a critical role in streamlining rescue operations, enhancing situational awareness, and restoring essential services in the early hours following a disaster.

6 Conclusion

The Smart Modular Workstation offers a comprehensive, integrated solution for disaster relief operations. With its modular design, sustainable power architecture, reliable communication systems, and robust engineering validated through calculations and simulations, it is well-equipped to support critical field operations and expedite response times in emergency scenarios. This project exemplifies interdisciplinary collaboration and innovation in emergency response engineering.